

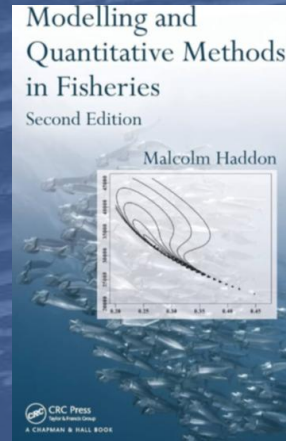
ESM 21: Applied Population Ecology

Modelling and Quantitative Methods in Fisheries (MQMF)

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What does the package do?

- Incorporates functions for Fisheries Stock Assessments, Ecological Modeling, and Parameter Estimate
- Provides functions and datasets that facilitate the implementation of various quantitative methods discussed in the book



What problems is the package best suited to address?

1. `bce()`: The Baranov Catch Equation
2. `bh()`: The Beverton-Holt model describes the relationship between spawning stock size and the number of young fish that successfully recruit into the population.
3. `discretelogistic()`: A basic population model used in ecology to describe how populations grow over time

Model Overview

Baranov Catch Equation (Baranov, 1918)

- Calculates fish catch by considering both fishing and natural mortality.
- Tracks how different age groups (cohorts) are affected over time.
- Helps estimate total fish removed by fishing **while accounting for natural mortality**.
- Used in stock assessments to predict catch levels.

Model Overview

Baranov Catch Equation

$$\hat{C}_{a,t} = \frac{F_{a,t}}{M + F_{a,t}} N_{a,t} \left(1 - e^{-(M + F_{a,t})} \right)$$

(Xiao, 2005)

$C_{a,t}$: Expected catch of fish of age a in year t .

$F_{a,t}$: Fishing mortality specific to age a .

M : Natural mortality rate (assumed constant).

$N_{a,t}$: the number of fish of age a alive at the beginning of year t

What data is needed?

- **M** → natural mortality: the proportion of the fish population that dies due to natural causes (assumed constant)
- **Ft** → The fishing mortality rate from a harvest rate (e.g., 0.2, or 20%)
- **Fat** → This is a vector representing the rate at which fish are removed from the population due to fishing, and it varies by age
- **Nt** → The population size of fish at each age at the beginning of a specific time period
- **Age** → This defines the age structure of the population being modeled, typically ranging from age 0 (new recruits) to some maximum age (e.g., the oldest individuals in the population).

Use Case

Setting Catch Limits

- In order to determine the maximum sustainable yield, the remaining population and yearly catch is needed.

Determining Population

- A simplified BCE can also be used to estimate the population if only the catches are reported, using algebra.

$$C = F * N \quad \longrightarrow \quad N = C / F$$



Example: California Halibut

- Taken from MQMF package documentation
- Applying real world analysis to California halibut



```
{r}  
  
# Age range of California halibut  
#(California halibut can live up to 30 years, individuals older than 15 years are rare)  
age_range <- 0:30  
  
# 22% (0.22) is a hypothetical value  
harvest_rate = 0.22  
# Compute the instantaneous fishing mortality (Ft = -ln(1 - harvest_rate))  
# Ft ≈ 0.2231 or fishing mortality is about 22.3% per year  
instantaneous_fishing_mortality <- -log(1 - harvest_rate)  
  
# Fishing mortality (F) at age (n)  
fishing_mortality_n <- rep(instantaneous_fishing_mortality, length(age_range))  
  
# Studies have estimated the natural mortality rate (M) for California halibut 20% per year.  
# Natural Mortality (M)  
M <- 0.20  
  
# Initial population size of 1000 is a hypothetical value  
# Actual population depend on stock assessments and environmental conditions  
# Initial Population (Nt)  
Nt <- 1000
```

Example: Results

bce() function produces a data frame:

1. surviving numbers-at age (Nt)
2. total mortality-at-age (N-Dying)
3. Catch-at-age (Catch)

What is the population and catch of California Halibut at 10 years old?

	Nt	N-Dying	Catch
0	1000.0000000	NA	NA
1	709.5363494	101.57742421	188.88622642
2	503.4418311	72.07287475	134.02164354
3	357.2102789	51.13832444	95.09322770
4	253.4536773	36.28450004	67.47210163
5	179.8345969	25.74517169	47.87390867
6	127.5991834	18.26713514	33.96827839
7	90.5362588	12.96119638	24.10172824
8	64.2387665	9.19643996	17.10105227
9	45.5797399	6.52520844	12.13381820
10	32.3404822	4.62987257	8.60938507
11	22.9467477	3.28506288	6.10867165
12	16.2815516	2.33087153	4.33432458
13	11.5523527	1.65383807	3.07536084
14	8.1968142	1.17345823	2.18208030
15	5.8159376	0.83261127	1.54826529
16	4.1266191	0.59076796	1.09855050
17	2.9279863	0.41917134	0.77946151
18	2.0775127	0.29741730	0.55305628
19	1.4740708	0.21102839	0.39241353
20	1.0459068	0.14973231	0.27843166
21	0.7421089	0.10624052	0.19755739
22	0.5265532	0.07538151	0.14017415
23	0.3736087	0.05348592	0.09945865
24	0.2650889	0.03795021	0.07056953
25	0.1880902	0.02692705	0.05007165

Portfolio Assignment



Instructions:

- The MQMF_Portfolio_Assignment.qmd uses the example numbers for California halibut. For your assignment please choose a fish species you're interested in:
 - **Maximum Age** (As a series of numbers, something like 0:20)
 - **Natural mortality**
 - **Initial population size**
 - If you can't find a stock assessment for your **species and location**, just use 1000
- Once you've filled in this information, **use the bce() method to model catch for your population, analyze the data, and answer the conservation focused research questions.**

Advantages

- Includes a dedicated book for guidance
- Incorporates multiple functions, models, and methods for assessing fish populations, making it useful for different fishery contexts
- Good integration with R

Limitations

- Primarily used alongside the book, limiting usability
- Less widely adopted compared to packages like FSA
- Potentially a steeper learning curve without familiarity with the book or background in fishery
- Potentially less compatible with other packages

Resources for MQMF package

1. [Cran R](#)

1. [MQMF book](#)

2. [Package Repository](#)

1. [R documentation](#)

Interested in studying Fish Stocks?

- [Simple Fisheries Stock Assessment \(FSA\)](#)
- [Stock Assessment Model Descriptions](#)
- [Fish Data](#)

